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By Amit Katiyar (MCA-JNU)



1. Let $f(x) = [x^2 - 3]$, where $[\]$ denotes the greatest integer function. Then the number of points in the interval $(1, 2)$, where the function is discontinuous, is

NIMCET 2009

- (a) 4 (b) 2
(c) 6 (d) None

2. If $f(x) = \begin{cases} x \sin\left(\frac{1}{x}\right) & \text{for } x \neq 0 \\ 0 & \text{for } x = 0 \end{cases}$, then

NIMCET 2010

- (a) f is continuous functions
(b) $f'(0^+)$ exist but $f'(0^-)$ does not exist
(c) $f(0^+) \neq f(0^-)$
(d) $f'(0^+)$ and $f'(0^-)$ do not exist

3. What is the value of a for which

NIMCET 2012

$$f(x) = \begin{cases} \sin x, & \text{if } x \leq \frac{\pi}{2} \\ ax, & \text{if } x < \frac{\pi}{2} \end{cases} \text{ is continuous?}$$

- (a) π (b) $\frac{\pi}{2}$
(c) $\frac{2}{\pi}$ (d) 0

4. Consider the function f defined by $f(x) = \begin{cases} x^2 - 1, & x < 3 \\ 2ax, & x \geq 3 \end{cases}$ for all real number x . if f is continuous at $x = 3$, then value a is

NIMCET 2016

- (a) 8 (b) $\frac{3}{4}$ (c) $\frac{1}{8}$ (d) $\frac{4}{3}$

5. The slope of the function $f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right), & x \neq 0 \\ 0, & x = 0 \end{cases}$

NIMCET 2017

- (a) 1 (b) 0 (c) -1 (d) None

6. $f(x) = x + |x|$ continuous for

NIMCET 2018

- (a) $x \in (-\infty, \infty)$ (b) $x \in (-\infty, \infty) - \{0\}$
(c) only $x > 0$ (d) No value of x

7. The set of points, where $f(x) = \frac{x}{1+|x|}$ is differentiable is

NIMCET 2018

- (a) $(-\infty, -1) \cup (-1, \infty)$ (b) $(-\infty, \infty)$
(c) $(0, \infty)$ (d) $(-\infty, 0) \cup (0, \infty)$

8. Let $f : R \rightarrow R$ be defined by

NIMCET 2019

$$f(x) = \begin{cases} x \sin\left(\frac{1}{x}\right), & \text{if } x > 0 \\ 0, & \text{if } x < 0 \end{cases} \text{ Then,}$$

- (a) f is neither continuous nor differentiable at 0
(b) f is continuous and differentiable at 0
(c) f is continuous but not differentiable at 0
(d) f is not continuous but differentiable at 0

9. Let $f(x) = \begin{cases} \cos[x], & x \geq 0 \\ |x| + a, & x < 0 \end{cases}$ where $[x]$ denotes the greatest integer $\leq x$. If f should be continuous at $x = 0$, then a must be

NIMCET 2019

- (a) 0 (b) 1
(c) 2 (d) -1

10. If $f(x) = \begin{cases} x^2, & x \leq 0 \\ 2 \sin x, & x > 0 \end{cases}$, then $x = 0$ is

NIMCET 2020

- (a) point of minima (b) point of maxima
(c) point of discontinuity (d) None of the above

11. If $g(x) = \begin{cases} (x^2 - x)/2x, & x \neq 0 \\ k, & x = 0 \end{cases}$ is a continuous function at $x = 0$, then the value of k is

NIMCET 2020

- (a) 2 (b) $1/2$ (c) 1 (d) None of these

12. Test the continuity of the function at $x = 2$

$$f(x) = \begin{cases} \frac{5}{2} - x, & x < 2 \\ 1, & x = 2 \\ x - \frac{3}{2}, & x > 2 \end{cases}$$

NIMCET 2020

- (a) Continuous at $x = 2$
(b) Discontinuous at $x = 2$
(c) Semicontinuous at $x = 2$
(d) None of the above

13. If $f : R \rightarrow R$ is defined by

$$f(x) = \begin{cases} \frac{x+2}{x^2+3x+2} & \text{if } x \in R - \{-1, -2\} \\ -1 & \text{if } x = -2 \\ 0 & \text{if } x = -1 \end{cases} \text{ , then } f(x)$$

is continuous on the set

NIMCET 2021

- (a) R (b) $R - \{-2\}$
(c) $R - \{-1\}$ (d) $R - \{-1, -2\}$

14. The function $f(x) = \begin{cases} (1 + 2x)^{1/x}, & x \neq 0 \\ e^2, & x = 0 \end{cases}$ is

NIMCET 2022

- (a) differentiable at $x = 0$
(b) continuous at $x = 0$
(c) discontinuous at $x = 0$
(d) continuous but not differentiable at $x = 0$

15. $f(x) = x + |x|$ continuous for

NIMCET 2022

- (a) $x \in (-\infty, \infty)$ (b) $x \in (-\infty, \infty) - \{0\}$
(c) only $x > 0$ (d) No value of x

16. Number of points of which $f(x)$ is not differentiable $f(x) = |\cos x| + 3$ in $[-\pi, \pi]$

NIMCET 2023

- (a) 2 (b) 3
(c) 4 (d) None of the above

17. Let $f : R \rightarrow R$ be a function such that $f(0) = \frac{1}{\pi}$ and $f(x) = \frac{x}{e^{\pi x} - 1}$ for $x \neq 0$ (Continuity and differentiability)

- (a) $f(x)$ is not continuous at $x = 0$
(b) $f(x)$ is continuous but not differentiable at $x = 0$
(c) $f(x)$ is differentiable at $x = 0$ and $f'(0) = -\frac{\pi}{2}$
(d) None of these

NIMCET 2024

18. Let $f(x) = \begin{cases} x^2 \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$

Then which of the following is true

- (a) $f(x)$ is not continuous at $x = 0$
(b) $f(x)$ is not differentiable at $x = 0$
(c) $f'(x)$ is not continuous at $x = 0$
(d) $f'(x)$ is continuous at $x = 0$

NIMCET 2024



19. Let $f: \mathbb{R} \rightarrow \mathbb{R}$, be any function defined as $f(x) = \begin{cases} x^\alpha \sin \frac{1}{x^\beta}, & x \neq 0 \\ 0, & x = 0 \end{cases}$ $\alpha, \beta \in \mathbb{R}$, which of the following is true? ($\mathbb{R}$ denotes the set of all real numbers)

NIMCET 2025

- (a) $f(x)$ is continuous at $x = 0$, for all α & $\beta \in \mathbb{R}$
 (b) $f(x)$ is differentiable at $x = 0$ for all $\alpha > 0$ & $\beta > 0$
 (c) $f(x)$ is continuous at $x = 10$ for only $\alpha > 0$ & $\beta > 0$
 (d) $f(x)$ is continuous at $x = 0$ for all $\alpha > 0$ & $\beta \in \mathbb{R}$

ANSWER KEY

1.	b	2.	a	3.	c	4.	d	5.	b
6.	a	7.	b	8.	c	9.	b	10.	a
11.	d	12.	b	13.	c	14.	b	15.	a
16.	a	17.	b	18.	c	19.	d		

ANALYSIS 2007 TO 2025

QUESTIONS ASKED IN NIMCET 2007 TO 2025

Years	No. of Questions	Years	No. of Questions
NIMCET 2007	00	NIMCET 2016	01
NIMCET 2008	00	NIMCET 2017	01
NIMCET 2009	01	NIMCET 2018	02
NIMCET 2010	01	NIMCET 2019	02
NIMCET 2011	00	NIMCET 2020	03
NIMCET 2012	01	NIMCET 2021	01
NIMCET 2013	00	NIMCET 2022	02
NIMCET 2014	00	NIMCET 2023	01
NIMCET 2015	00	NIMCET 2024	02
		NIMCET 2025	01

SOLUTION

1. (b) $y = [f(x)]$ is discontinuous, where $f(x)$ is an integer,

$$f(x) = [x^2 - 3], x \in (1, 2)$$

$$1 < x < 2 \Rightarrow 1 < x^2 < 4$$

$$\Rightarrow -2 < x^2 - 3 < 1$$

Here, the value of $f(x)$, where $(x^2 - 3)$ is an integer

$$f(x) = \begin{cases} -2 & 1 < x < \sqrt{2} \\ -1 & \sqrt{2} < x < \sqrt{3} \\ 0 & \sqrt{3} < x < 2 \end{cases}$$

Two points $\sqrt{2}, \sqrt{3}$ in the interval $(1, 2)$ for which given function is discontinuous.

2. (a) $f(x) = \begin{cases} x \sin \left(\frac{1}{x} \right), & x \neq 0 \\ 0, & x = 0 \end{cases}$

$$\begin{aligned} \text{LHL} &= \lim_{x \rightarrow 0^-} x \sin \left(\frac{1}{x} \right) = \lim_{x \rightarrow 0^-} \frac{\sin \left(\frac{1}{x} \right)}{\left(\frac{1}{x} \right)} \\ &= \frac{\sin \left(\frac{1}{0^-} \right)}{\left(\frac{1}{0^-} \right)} \end{aligned}$$

$$= \frac{1 \leq \sin(-\infty) \leq 1}{-\infty} = 0$$

$$\text{RHL} = \lim_{x \rightarrow 0^+} x \sin \left(\frac{1}{x} \right)$$

$$= \lim_{x \rightarrow 0^+} \frac{\sin \left(\frac{1}{x} \right)}{\left(\frac{1}{x} \right)}$$

$$= \frac{\sin \left(\frac{1}{0^+} \right)}{\left(\frac{1}{0^+} \right)}$$

$$= \frac{-1 \leq \sin(\infty) \leq 1}{\infty} = 0$$

Hence, $\text{LHL} = \text{RHL} = f(0)$ function is continuous at $x = 0$.

3. (c)

$$f(x) = \begin{cases} \sin x, & x \leq \frac{\pi}{2} \\ ax, & x > \frac{\pi}{2} \end{cases}$$

$$\begin{aligned} \text{LHL} &= \lim_{x \rightarrow \left(\frac{\pi}{2} \right)^-} \sin x \\ &= \sin \frac{\pi}{2} = 1 \end{aligned}$$

$$\text{RHL} = \lim_{x \rightarrow \left(\frac{\pi}{2} \right)^+} ax = a \left(\frac{\pi}{2} \right)$$

Function is continuous at $x = \frac{\pi}{2}$.

Hence, $\text{LHL} = \text{RHL}$

$$\Rightarrow a \times \frac{\pi}{2} = 1 \Rightarrow a = \frac{2}{\pi}$$

4. (d) $\text{LHL} = \lim_{x \rightarrow 3^-} x^2 - 1$
 $= 9 - 1 = 8$

$$\text{RHL} = \lim_{x \rightarrow 3^+} 2a \cdot 3 = 6a$$

Function is $x \rightarrow 3^+$ continuous at $x = 3$

Hence, $\text{LHL} = \text{RHL}$

$$\Rightarrow 8 = 6a$$

$$\Rightarrow a = \frac{4}{3}$$

5. (b) $f(x) = x^2 \sin \frac{1}{x}, f(0) = 0$

$$\begin{aligned} \text{LHD } f'(x) &= \lim_{h \rightarrow 0} \frac{f(0-h) - f(0)}{-h} \\ &= \frac{(-h)^2 \sin \left(\frac{1}{-h} \right)}{(-h)} = \lim_{h \rightarrow 0} \frac{\sin \left(\frac{1}{h} \right)}{\left(\frac{1}{h} \right)} \end{aligned}$$

$$= \frac{-1 \leq \sin(\infty) \leq 1}{\infty} = 0$$

$$\text{RHD } f'(x) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h}$$

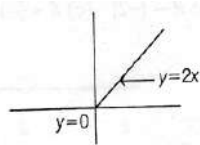
$$= \frac{h^2 \sin \left(\frac{1}{h} \right)}{h} = \frac{\sin \left(\frac{1}{h} \right)}{\left(\frac{1}{h} \right)}$$

$$= \frac{-1 \leq \sin(\infty) \leq 1}{\infty} = 0$$

$\text{LHD} = \text{RHD} = f'(0) = 0 = \text{Slope of function.}$



6. (a)



$$f(x) = x + |x| = \begin{cases} 2x & x \geq 0 \\ 0 & x < 0 \end{cases}$$

Continuous everywhere.

7. (b) $f(x) = \frac{x}{1+|x|} = \begin{cases} \frac{x}{1-x}, & x < 0 \\ \frac{x}{1+x}, & x \geq 0 \end{cases}$

$$\text{LHD } f'(x) = \lim_{x \rightarrow 0^-} \frac{f(x) - f(0)}{x - 0}$$

$$= \lim_{x \rightarrow 0^-} \left(\frac{x}{1-x} \right) \frac{1}{x}$$

$$= \lim_{x \rightarrow 0^-} \frac{1}{1-x} = 1$$

$$\text{RHD } f'(x) = \lim_{x \rightarrow 0^+} \frac{f(x) - f(0)}{x - 0}$$

$$= \lim_{x \rightarrow 0^+} \left(\frac{x}{1+x} \right) \frac{1}{x}$$

$$= \lim_{x \rightarrow 0^+} \frac{1}{1+x} = 1$$

Hence, LHD = RHD

So, it is differentiable everywhere.

8. (c) For continuity of $f(x)$

$$\text{LHL} = \lim_{x \rightarrow 0^-} x \sin\left(\frac{1}{x}\right)$$

$$= \lim_{x \rightarrow 0^-} \frac{\sin\left(\frac{1}{x}\right)}{\left(\frac{1}{x}\right)} = \frac{\sin\left(\frac{1}{0^-}\right)}{\left(\frac{1}{0^-}\right)}$$

$$= \frac{-1 \leq \sin(-\infty) \leq 1}{-\infty} = 0$$

$$\text{RHL} = \lim_{x \rightarrow 0^+} x \sin\left(\frac{1}{x}\right)$$

$$= \lim_{x \rightarrow 0^+} \frac{\sin\left(\frac{1}{x}\right)}{\left(\frac{1}{x}\right)}$$

$$= \frac{\sin\left(\frac{1}{0^+}\right)}{\left(\frac{1}{0^+}\right)}$$

$$= \frac{-1 \leq \sin(\infty) \leq 1}{\infty} = 0$$

Hence, LHL = RHL = $f(0)$

Function is continuous at $x = 0$

For differentiability:

$$\text{LHD} = \lim_{h \rightarrow 0} \frac{f(0-h) - f(0)}{(-h)}$$

$$= \lim_{h \rightarrow 0} \frac{(-h) \sin\left(\frac{1}{-h}\right)}{-h}$$

$$= \sin(-\infty) = \text{DNE (Does Not Exist)}$$

$$\text{as } -1 \leq \sin(-\infty) \leq 1$$

$$\text{RHD} = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{(h)}$$

$$= \frac{h \sin\left(\frac{1}{h}\right)}{h}$$

$$= \sin(\infty) = \text{DNE (Does not exist)}$$

$$\text{As } -1 \leq \sin(\infty) \leq 1$$

Not differentiable at $x = 0$

9. (b) As function is continuous at $x = 0$

Its LHL = RHL = $f(0)$

$$\text{RHL} = \lim_{x \rightarrow 0^+} \cos[x] = \cos[0^+] = \cos 0 = 1$$

$$\text{LHL} = \lim_{x \rightarrow 0^-} |x| + a = 0 + a = a$$

$$\text{LHL} = \text{RHL}$$

$$a = 1$$

10. (a) Continuity:

$$f(x) = \begin{cases} x^2; & x \leq 0 \\ 2 \sin x; & x > 0 \end{cases}$$

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} x^2 = 0^2 = 0.$$

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} 2 \sin x = 2 \sin 0 = 0.$$

$$f(0) = 0^2 = 0$$

$\therefore \lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} f(0)$, the function $f(x)$ is continuous at $x = 0$.

Differentiability:

$$f(x) = \begin{cases} x^2; & x \leq 0 \\ 2 \cos x; & x > 0 \end{cases}$$

$$\lim_{x \rightarrow 0^-} f'(x) = \lim_{x \rightarrow 0^-} 2x = 2 \times 0 = 0$$

$$\lim_{x \rightarrow 0^+} f'(x) = \lim_{x \rightarrow 0^+} 2 \cos x = 2 \cos 0 = 2.$$

$\therefore \lim_{x \rightarrow 0^+} f'(x) \neq \lim_{x \rightarrow 0^-} f'(x)$, the function is not differentiable at $x = 0$.

Since, the function is not differentiable at $x = 0$, let us

examine the possibility of maximum/ minimum at the point.

The function $f(x) = x^2$ is strictly decreasing in $(-\infty]$ and its minimum is $0^2 = 0$ at $x = 0$.

The function $f(x) = 2 \sin x$ is strictly increasing in $(0, \frac{\pi}{2}]$ and its minimum is $2 \sin 0 = 0$ at $x = 0$.

So the function $f(x)$ has a local minimum at $x = 0$.

11. (d)

As function is continuous at $x = 0$

Its LHL = RHL = $g(0)$

$$\text{LHL} = \text{RHL} = \lim_{x \rightarrow 0} \frac{x^2 - x}{2x}$$

$$= \lim_{x \rightarrow 0} \left(\frac{x}{2} - \frac{1}{2} \right)$$

$$= -\frac{1}{2}$$

$$\text{LHL} = \text{RHL} = g(0)$$

$$\Rightarrow -\frac{1}{2} = k$$

12. (b) $\text{LHL} = \frac{5}{2} - 2 = \frac{1}{2}$

$$\text{RHL} = 2 - \frac{3}{2} = \frac{1}{2}, f(2) = 1,$$

$$\text{LHL} \neq f(2) \neq \text{RHL}$$



Function is discontinuous at $x = 2$.

$$\begin{aligned} 13. (c) f(x) &= \frac{x+2}{x^2+2x+x+2} \\ &= \frac{(x+2)}{x(x+2)+1(x+2)} = \frac{(x+2)}{(x+2)(x+1)} \\ &= \frac{1}{(x+1)} \end{aligned}$$

Discontinuous at $x = -1$

Continuous at $\mathbb{R} - \{-1\}$

$$14. (b) \text{ The function } f(x) = \begin{cases} (1+2x)^{1/x}, & x \neq 0 \\ e^2, & x = 0 \end{cases}$$

Check continuity

at $x = 0$

$$\text{LHL} = \text{RHL} = f(0)$$

$$\Rightarrow \lim_{x \rightarrow 0} (1+2x)^{1/x} = e^2$$

$$\Rightarrow e^{\lim_{x \rightarrow 0} (1+2x-1) \times \frac{1}{x}} = e^2$$

$$\left[\begin{array}{l} \text{Use formula } \lim_{x \rightarrow a} (1+f(x))^{g(x)} \\ \lim_{x \rightarrow 0} (1+f(x)-1)g(x) \\ \text{then, } e^{x \rightarrow 0} \end{array} \right]$$

$$\Rightarrow e^{\lim_{x \rightarrow 0} \frac{2x}{x}} = e^2$$

$$\Rightarrow e^2 = e^2$$

$f(x)$ is continuous at $x = 0$.

For differentiability:

$$f(x) = (1+2x)^{1/x}$$

$$\log f(x) = \frac{1}{x} \log(1+2x)$$

Differentiate both side

$$\frac{1}{f(x)} f'(x) = \frac{2}{x(1+2x)} + \log(1+2x) \left(\frac{-1}{x^2} \right)$$

$$f'(x) = (1+2x)^{1/x} \left[\frac{2}{x(1+2x)} - \frac{\log(1+2x)}{x^2} \right]$$

$f'(x)$ is undefined at $x = 0$.

$\Rightarrow f(x)$ is not differentiable at $x = 0$

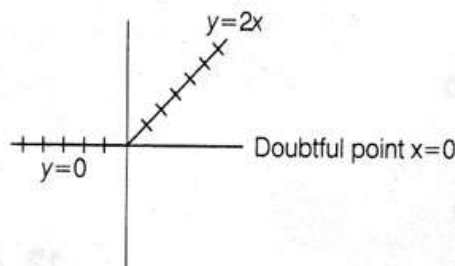
$$15. (a) f(x) = x + |x| \text{ is continuous for}$$

$$f(x) = \begin{cases} x+x=2x, & x \geq 0 \\ x-x=0, & x < 0 \end{cases}$$

$$\text{LHL} = \text{RHL} = f(0)$$

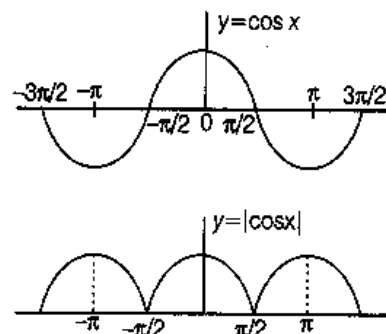
$$\lim_{x \rightarrow 0^-} = \lim_{x \rightarrow 0^+} 2x = f(0)$$

Hence, continuous at $x = 0$



Continuous at $x \in (-\infty, \infty)$

$$16. (a)$$



It is not differentiable at two points (We should know the function is not differentiable where there is sharp turns).

$y = |\cos x| + 3$ is not differentiable at

$$x = \frac{-\pi}{2}, \frac{\pi}{2}$$

$$17. (b) f(0) = \frac{1}{\pi}$$

$$f(x) = \frac{x}{e^{\pi x} - 1} : x \neq 0$$

Solution: Continuity

$$\text{LHL } \lim_{h \rightarrow 0} \frac{(0-h)}{e^{\pi(0-h)} - 1} = \lim_{h \rightarrow 0} \frac{-h}{e^{-\pi h} - 1} = 0$$

$$\text{RHL } \rightarrow \lim_{h \rightarrow 0} \frac{(0+h)}{e^{\pi(0+h)} - 1} = \lim_{h \rightarrow 0} \frac{h}{e^{\pi h} - 1} = 0$$

LHL to RHL continuous at $x = 0$

Differentiable:

$$\text{LHD } f'(x) = \lim_{h \rightarrow 0} \frac{f(0-h) - f(0)}{-h}$$

$$= \lim_{h \rightarrow 0} \left(\frac{-h}{e^{\pi h} - 1} - \frac{1}{\pi} \right) \left(\frac{-1}{h} \right)$$

$$= \lim_{h \rightarrow 0} \left(\frac{1}{e^{\pi h} - 1} + \frac{1}{\pi h} \right)$$

= does not exist.

$$\text{RHD } f'(x) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h}$$

$$f'(x) \lim_{h \rightarrow 0} \left(\frac{h}{e^{\pi h} - 1} - \frac{1}{\pi} \right) \frac{1}{h} = \lim_{h \rightarrow 0} \left(\frac{1}{e^{\pi h} - 1} - \frac{1}{\pi h} \right)$$

Does not exist.

Function is continuous but not diff. at $x = 0$

$$18. (c) f(x) = \begin{cases} x^2 \sin \frac{1}{x} & : x \neq 0 \\ 0 & : x = 0 \end{cases}$$

Continuity:

$$\lim_{h \rightarrow 0} (0-h)^2 \sin \frac{1}{0-h}$$

$$= \lim_{h \rightarrow 0} h^2 \sin \frac{-1}{h}$$



= 0 x oscillating

= 0

RHL

$$\lim_{h \rightarrow 0} (0+h)^2 \sin \frac{1}{0+h}$$

$$\lim_{h \rightarrow 0} h^2 \sin \frac{1}{h}$$

= 0 x oscillating

= 0

LHL = RHL Contⁿ at x = 0

Differentiable:

LHD

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(0-h)-f(0)}{-h}$$

$$= \lim_{h \rightarrow 0} \frac{h^2 \sin\left(\frac{-1}{h}\right) - 0}{-h}$$

$$= \lim_{h \rightarrow 0} \frac{-h^2 \sin \frac{1}{h}}{-h}$$

$$= \lim_{h \rightarrow 0} \frac{\sin 1/h}{1/h}$$

RHD:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(0+h)-f(0)}{h}$$

$$\lim_{h \rightarrow 0} \frac{h^2 \sin \frac{1}{h} - 0}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\sin 1/h}{1/h}$$

Does not exist.

Function is continuous but not diff. at x = 0.

19. (d)

For x ≠ 0,

$$f(x) = x^\alpha \sin\left(\frac{1}{x^\beta}\right), \quad f(0) = 0$$

we have

$$|f(x)| \leq |x|^\alpha$$

As x → 0:

$$\text{- If } \alpha > 0, |x|^\alpha \rightarrow 0 \Rightarrow \lim_{x \rightarrow 0} f(x) = 0 = f(0) \Rightarrow$$

continuous

$$\text{- If } \alpha \leq 0, \text{ the limit doesn't go to } 0 \Rightarrow \text{not continuous}$$

Thus f is continuous at x = 0 iff α > 0, for any β